Chapter 7. The Bootstrap

Sampling distributions, bootstrap estimates of standard error and bias, resampling schemes

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7.1 The plug-in principle

Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ and $\hat{\theta} = T(X_1, \dots, X_n)$ estimate $\theta = \theta(F)$. The **sampling distribution** G_F of $\hat{\theta}$ (bias, standard error, quantiles) depends on the unknown F. The bootstrap replaces F by an estimate $\hat{F}$ and simulates:

$$\text{real world: } F \to \mathbb{X}_n \to \hat{\theta}, \qquad \text{bootstrap world: } \hat{F} \to \mathbb{X}_n^* \to \hat{\theta}^*.$$

The distribution of $\hat{\theta}^*$ given the data, $G^* = G_{\hat{F}}$, is the **bootstrap distribution**; it estimates G_F.

Nonparametric bootstrap. $\hat{F} = F_n$ (mass $1/n$ at each observation): draw n observations *with replacement*.

1. Draw a bootstrap sample $X_1^*, \dots, X_n^*$ with replacement from $\{X_1, \dots, X_n\}$ and compute $\hat{\theta}^* = T(X_1^*, \dots, X_n^*)$.
2. Repeat B times to get $\hat{\theta}^{*1}, \dots, \hat{\theta}^{*B}$; their empirical distribution approximates G^*.

Two errors: $F_n \neq F$ (statistical, $O(n^{-1/2})$) and $B < \infty$ (Monte Carlo, $O(B^{-1/2})$, reducible at will).

7.2 Bootstrap estimates of standard error and bias

Replacing F by F_n in $\text{se}_F(\hat{\theta})$ and $\text{bias}_F(\hat{\theta}) = E_F\hat{\theta} - \theta(F)$ gives

$$\widehat{\text{se}}_{\text{Boot}} = \Big(\frac{1}{B-1}\sum_b (\hat{\theta}^{*b} - \bar{\theta}^*)^2\Big)^{1/2}, \qquad \widehat{\text{bias}}_{\text{Boot}} = \bar{\theta}^* - \hat{\theta}, \qquad \bar{\theta}^* = \frac{1}{B}\sum_b \hat{\theta}^{*b}.$$

The true $\theta(F)$ is replaced by $\theta(F_n)$, which is $\hat{\theta}$ for plug-in statistics.

Example 7.1 (An exact bootstrap distribution)

Data $\{1, 4, 9\}$, $\hat{\theta} = \text{median} = 4$. Of the 27 equally likely resamples, the median is 1 iff at least two draws are 1: probability $(3 \cdot 2 + 1)/27 = 7/27$; likewise 9 w.p. $7/27$ and 4 w.p. $13/27$. Hence

$$E^*\hat{\theta}^* = \frac{7 \cdot 1 + 13 \cdot 4 + 7 \cdot 9}{27} = 4.52, \qquad \text{se}^* = \Big(\frac{7 + 13 \cdot 16 + 7 \cdot 81}{27} - 4.52^2\Big)^{1/2} = 2.92,$$

and the bias estimate is 0.52, computed exactly since n is tiny.

When the bootstrap works

For smooth statistics (smooth functions of means, many M- and U-statistics) the bootstrap law of $\sqrt{n}(\hat{\theta}^* - \hat{\theta})$ consistently estimates that of $\sqrt{n}(\hat{\theta} - \theta)$. It can fail at the boundary: for $\hat{\theta} = \max_i X_i$, Uniform$(0, \theta)$, $P^*(\hat{\theta}^* = \hat{\theta}) = 1 - (1 - 1/n)^n \to 1 - e^{-1}$ but $P(\hat{\theta} = \theta) = 0$.

7.3 Parametric bootstrap

If $F = F_\eta$ is parametric, estimate $\hat{\eta}$ (e.g. by ML) and resample from $F_{\hat{\eta}}$. More accurate when the model is right and useful for intractable sampling laws (e.g. nonlinear functionals of a Weibull fit); badly biased when it is wrong.

7.4 Resampling designs

Rule: *resample the way the data were generated.*

- **Several samples.** Resample within each sample, keeping $n_1, \ldots, n_k$.
- **Random design.** If $(X_i, Y_i) \overset{\text{i.i.d.}}{\sim} F$, resample pairs (**case** bootstrap); valid even if the regression model is wrong.
- **Fixed design.** If $Y_i = \alpha + \beta x_i + \epsilon_i$ with i.i.d. errors, set $Y_i^* = \hat{\alpha} + \hat{\beta} x_i + r_i^*$, r_i^* drawn from the (leverage-adjusted) residuals; more efficient but relies on the model, including homoscedasticity.

Choosing:

- Classical inference is conditional on the x's; the residual bootstrap reproduces this and matches normal theory when it is adequate.
- The case bootstrap stays valid for a wrong linear model (β = minimizer of $E(Y - \alpha - \beta X)^2$) and heteroscedasticity, at some loss of efficiency.
- Designed experiments (x's chosen) argue for fixed x.

The bootstrap samples *with* replacement and keeps the structure of the alternative (pairs stay together); a permutation test samples *without* replacement and destroys the association to produce the null.

7.5 Selected exercises

Motivation. Ideal bootstrap ($B = \infty$) computations by hand, the uniform-maximum failure, and counting resamples.

Exercise 7.1 (The empirical distribution function)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$ and $F_n(x) = n^{-1} \sum_i I(X_i \le x)$. For x with $F(x) \in (0, 1)$, show:
(a) $nF_n(x) \sim B(n, F(x))$;
(b) $F_n(x)$ is unbiased for $F(x)$ with standard error $\sqrt{F(x)(1 - F(x))/n}$.

Solution.

(a) The $I(X_i \le x)$ are independent Bernoulli($F(x)$), so $nF_n(x) \sim B(n, F(x))$.

(b) From (a), $E\, nF_n(x) = nF(x)$ and Var $nF_n(x) = nF(x)(1 - F(x))$. Divide by n and n^2.

So F_n is within $O_P(n^{-1/2})$ of F at each x: the statistical error of Section 7.1. ■

Exercise 7.5 (Ideal bootstrap for the mean and the variance)

Let $x_1, \ldots, x_n$ be fixed data with mean $\bar{x}$, $\hat{\sigma}^2 = \frac{1}{n} \sum_i (x_i - \bar{x})^2$ and $s^2 = \frac{n}{n-1} \hat{\sigma}^2$, and let $X_1^*, \ldots, X_n^*$ be drawn with replacement.
(a) Show that $E^* \bar{X}^* = \bar{x}$ and $\text{Var}^* \bar{X}^* = \hat{\sigma}^2/n$.
(b) For $\theta = \mu^2$ and $\hat{\theta}_1 = \bar{X}^2$, show that the ideal bootstrap bias estimate is $\hat{\sigma}^2/n$. Compare it with the true bias, and find the bias of the corrected estimator $2\hat{\theta}_1 - \bar{\theta}^*$.
(c) For $\theta = \sigma^2$ and $\hat{\theta}_2 = s^2$, show that $\bar{\theta}^* - \hat{\theta}_2 = -s^2/n$, although s^2 is unbiased. Explain.

Solution [Supplement].

(a) The X_i^* are i.i.d. F_n: $E^*X_1^* = \bar{x}$, $\mathrm{Var}^* X_1^* = \hat{\sigma}^2$. Average n copies.

(b) $E^*(\bar{X}^{*2}) = \mathrm{Var}^* \bar{X}^* + (E^*\bar{X}^*)^2 = \hat{\sigma}^2/n + \bar{x}^2$, so $\bar{\theta}^* - \hat{\theta}_1 = \hat{\sigma}^2/n$; the true bias is $\mathrm{Var}\,\bar{X} = \sigma^2/n$. The corrected estimator is $\bar{X}^2 - \hat{\sigma}^2/n$, and with $E\hat{\sigma}^2 = \frac{n-1}{n}\sigma^2$,

$$E(\bar{X}^2 - \hat{\sigma}^2/n) = \mu^2 + \frac{\sigma^2}{n} - \frac{(n-1)\sigma^2}{n^2} = \mu^2 + \frac{\sigma^2}{n^2}.$$

The bias drops from $O(n^{-1})$ to $O(n^{-2})$.

(c) s^{*2} is unbiased for the variance $\hat{\sigma}^2$ of F_n, so $\bar{\theta}^* = \hat{\sigma}^2$ and $\bar{\theta}^* - s^2 = -s^2/n$. The bootstrap bias is $E_{F_n}\hat{\theta}^* - \theta(F_n)$ with $\theta(F_n) = \hat{\sigma}^2 \neq s^2$; correctly centered it is 0. The shortcut $\bar{\theta}^* - \hat{\theta}$ holds only for plug-in statistics (Section 7.2). ■

Exercise 7.6 (Failure for the uniform maximum)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathrm{Uniform}(0, \theta)$ and $\hat{\theta} = \max_i X_i$.
(a) Show that $n(\theta - \hat{\theta})/\theta \xrightarrow{d} \mathrm{Exp}(1)$.
(b) Show that $P^*(\hat{\theta}^* = \hat{\theta}) = 1 - (1 - 1/n)^n \to 1 - e^{-1}$.
(c) Conclude that the bootstrap distribution of $n(\hat{\theta} - \hat{\theta}^*)/\hat{\theta}$ does not converge to $\mathrm{Exp}(1)$.

Solution [Supplement].

(a) For $t \geq 0$ and $n > t$, $P(n(\theta - \hat{\theta})/\theta > t) = P(\hat{\theta} < \theta(1 - t/n)) = (1 - t/n)^n \to e^{-t}$.

(b) A.s. $x_{(n)}$ is a single point, and $\hat{\theta}^* = \hat{\theta}$ iff it is drawn at least once in n draws of probability $1/n$: $1 - (1 - 1/n)^n \to 1 - e^{-1} \approx 0.632$.

(c) Let $V^* = n(\hat{\theta} - \hat{\theta}^*)/\hat{\theta} \geq 0$. Convergence to the continuous $\mathrm{Exp}(1)$ would give $P^*(V^* \leq \frac{1}{2}) \to 1 - e^{-1/2} = 0.393$, but $P^*(V^* \leq \frac{1}{2}) \geq P^*(V^* = 0) \to 0.632$: the bootstrap law keeps an atom at 0. The cause: $\hat{\theta}$ sits at the boundary of the support, where F_n cannot mimic F. ■

Exercise 7.7 (Counting bootstrap samples)

Let the data $x_1, \ldots, x_n$ be distinct.
(a) Show that there are $\binom{2n-1}{n}$ distinct bootstrap samples (as multisets), and that the one in which x_i appears c_i times has probability $\frac{n!}{c_1! \cdots c_n!} n^{-n}$.
(b) Show that the expected number of distinct original observations in a bootstrap sample is $n\{1 - (1 - 1/n)^n\} \approx 0.632\, n$.

Solution [Supplement].

(a) A multiset is given by counts $c_i \geq 0$, $\sum c_i = n$: n stars and $n - 1$ bars, $\binom{2n-1}{n}$ arrangements. The n^n ordered draws are equally likely, $n!/(c_1! \cdots c_n!)$ of them with given counts.

(b) With $J_i = I(x_i \text{ appears})$, $EJ_i = 1 - (1 - 1/n)^n$; sum over i, and $(1 - 1/n)^n \to e^{-1}$.

For $n = 3$: 10 multisets among 27 draws (Example 7.1), e.g. $\{1, 4, 9\}$ w.p. 6/27, $\{1, 1, 1\}$ w.p. 1/27. Since $\binom{2n-1}{n} \approx 4^n/(2\sqrt{\pi n})$, enumeration is infeasible and Monte Carlo is used. ■